

Fault Code: 219

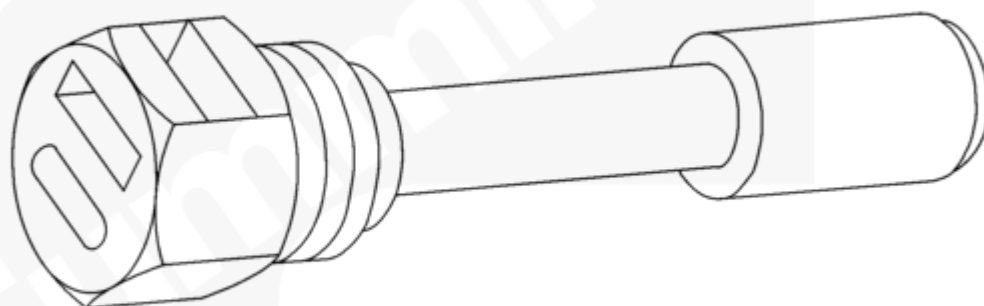
Make-Up Tank Oil Level - Low

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 219 PID(P): SPN: FMI: Lamp: Red SRT:	Make-Up Tank Oil Level - Low. This fault code is displayed to indicate that the make-up oil tank oil is below specified levels.	The Centinel™ system will not operate.

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Circuit Description

The make-up tank oil level low sensor monitors the oil level in the make-up oil tank and reports to the Centinel™ control module when the oil drops below the specified level.

Component Location

The make-up tank oil level low sensor is located on the lower portion of the front of the Centinel™ system make-up oil tank.

Shoptalk

Verify that the make-up tank has the appropriate level of oil.

Warnings and Cautions

⚠ CAUTION ⚠

To avoid pin and harness damage, use the following test leads when taking measurements:

Male Cannon, Metri-Pack, and Deutsch test lead, Part Number 3822758

Female AMP, Metri-Pack, and Deutsch test lead, Part Number 3822917

Male Deutsch test lead, Part Number 3823993

Female Deutsch test lead, Part Number 3823994

Male Weather-Pack test lead, Part Number 3823995

Female Weather-Pack test lead, Part Number 3823996.

⚠ CAUTION ⚠

The component can be damaged internally if dropped.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the make-up tank oil level low sensor.	
	STEP 1A. Inspect the make-up tank oil level low sensor and Centinel™ harness connectors.	No damaged pins
	STEP 1B. Check the make-up tank oil level low sensor for an open.	Less than 100 ohms
STEP 2.	Check the Centinel™ harness.	
	STEP 2A. Inspect the Centinel™ harness and the Centinel™ control module connectors.	No damaged pins
	STEP 2B. Check for an open circuit.	Less than 10 ohms

STEPS		SPECIFICATIONS
	STEP 2C. Check for a short circuit from pin to pin.	More than 1k ohms
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 219 inactive

STEP 1. Check the make-up tank oil level low sensor.

STEP 1A. Inspect the make-up tank oil level low sensor and the Centinel™ harness connector.

Conditions : - Fill the make-up oil tank. - Turn the keyswitch OFF. - Disconnect the Centinel™ harness from the make-up tank oil level low sensor.		
Action	Specification/Repair	Next Step

• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Wire insulation damage • Connector shell broken • Damaged locking tab connector. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	1B
	Repair the damaged pins Repair or replace the Centinel™ harness or the make-up tank oil level low sensor, whichever has damaged pins. • Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html). • Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the make-up tank oil level low sensor. Refer to Procedure 019-056 (/qs3/pubsys2/xml/en/procedures/96/96-019-056-tr.html). • Install the appropriate connector seal if damaged or missing.	3A

STEP 1B. Check the make-up tank oil level low sensor for an open.

Conditions :

- Fill the make-up oil tank.
- Turn the keyswitch OFF.
- Disconnect the make-up tank oil level low sensor from the Centinel™ harness.

Action	Specification/Repair	Next Step
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Measure the resistance from pin A to pin B of the make-up tank oil level low sensor connector while the sensor is covered with oil. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 100 ohms	2A
	Replace the make-up tank oil level low sensor Refer to Procedure 019-056 (/qs3/pubsys2/xml/en/procedures/96/96-019-056-tr.html).	3A

STEP 2. Check the Centinel™ wiring harness.

STEP 2A. Inspect the Centinel™ wiring harness and Centinel™ control module connectors.

Conditions : - Turn the keyswitch OFF. - Disconnect the Centinel™ harness from the Centinel™ control module.		
Action	Specification/Repair	Next Step

• Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Wire insulation damage • Connector shell broken • Damaged locking tab connector. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	2B
	Repair the damaged pins Repair or replace the Centinel™ harness or the Centinel™ control module, whichever has the damaged pins. • Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html). • Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). • Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	3A

STEP 2B. Check for an open circuit.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the make-up tank oil level low sensor from the Centinel™ harness.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step
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• Heavy-duty: Measure the resistance from pin 7 of the Centinel™ harness connector to pin A of the make-up tank oil level low sensor, harness side.	Less than 10 ohms	2C
• Heavy-duty: Measure the resistance from pin 3 of the Centinel™ harness connector to pin B of the make-up tank oil level low sensor, harness side. • High-horsepower: Measure the resistance from pin 13 of the Centinel™ harness connector to pin A of the make-up tank oil level low sensor, harness side. • High-horsepower: Measure the resistance from pin 26 of the Centinel™ harness connector to pin B of the make-up tank oil level low sensor, harness side. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Replace the Centinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	3A

STEP 2C. Check for a short circuit from pin to pin.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the make-up tank oil level low sensor.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step
• Heavy-duty: Measure the resistance from pins 3 and 1 of the Centinel™ harness connector to all adjacent pins in the connector. • High-horsepower: Measure the resistance from pins 13 and 22 of the Centinel™ harness connector to all adjacent pins in the connector. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms	3A
	Replace the Centinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault.

Conditions :

- Connect all the components.

Action	Specification/Repair	Next Step
• Start the engine and let it idle for at least 1 minute. • Verify that Fault Code 219 is inactive.	Fault Code 219 inactive	Complete
	Return to troubleshooting steps or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	1A

Last Modified: 25-Feb-2004
